Long Nguyen Khac

Hanoi, Vietnam | nk.long723@gmail.com | longnk0702.github.io | longnk0702 | longnk0702 | Google Scholar

Education

International School, Vietnam National University, Hanoi *Hanoi, Vietnam*
B.Eng. in Automation and Informatics *Sep. 2021 – Dec. 2026*

- **GPA:** 3.71/4.00; ranked in the top 4% of the cohort.
- Completed a one-year gap period for international research and exchange programs in Indonesia, Taiwan and Singapore (05/2025 – 05/2026).
- **Research interest:** robotics, wireless power transfer, and AI.

Research Experience

Water & Energy Research Lab, Nanyang Technological University *Singapore*
Research Fellow *Feb. 2026 – Mar. 2026*

- Conducted research on relay-coil-enhanced spatial wireless power transfer systems for autonomous underwater vehicle charging.
- Analyzed coil configurations, power transfer efficiency, and system-level integration for docking-free underwater charging.
- Explored mechanical and communication requirements for integrating wireless charging into an AUV prototype.

AISC Lab, AIM-HI Institute, National Chung Cheng University *Taiwan / Remote*
Research Intern *May 2025 – Present*

- Appling machine learning models, including Random Forest, XGBoost, and Support Vector Regression, to predict cutting forces in CNC milling from Taguchi machining datasets.
- Developed data-driven modeling approaches for tool wear & cutting force prediction in CNC milling using experimental datasets.
- Applied nonlinear system analysis, sparse identification, and machine learning methods to model machining dynamics.
- Built data processing and model evaluation pipelines for tool wear & cutting force prediction under different experimental conditions.

Intelligent Control and Artificial Intelligence Laboratory, VNUIS *Hanoi, Vietnam*
Student Researcher *Sep. 2022 – Present*

- Conducting research on adaptive control for AUVs, focusing on station keeping for planar wireless charging.
- Designed the mechanical model and physical prototype of a 3-DOF delta parallel robot, and implemented adaptive fuzzy backstepping control for robust trajectory tracking under uncertainties.
- Built a dataset of 2,632 construction-site images and trained YOLOv5s-based models for personal protective equipment detection.

Selected International Experience

Nanyang Technological University *Singapore*
Global Connect Fellow *Feb. 2026 – Mar. 2026*

- Selected as 1 of 39 young scholars worldwide and 1 of 5 outstanding Vietnamese young scholars for a fully funded in the Global Connect Fellowship program at NTU, Singapore (∼2% acceptance rate from ∼2,700 applicants).
- Joined an leading research environment with exposure to interdisciplinary work across energy systems, and robotics.

National University of Singapore *Singapore*
Exchange Student *Aug. 2025 – Dec. 2025*

- Selected as one of 8 outstanding VNU students receiving a fully funded scholarship DiscoverNUS & the NUS ASEAN Master's Scholarship Programme for one semester exchange in NUS.
- Completed a one-semester academic exchange, gaining broader exposure to engineering education, international research culture, and interdisciplinary learning.

National Chung Cheng University *Taiwan*
Research Intern *May 2025 – Jul. 2025*

- Selected as one of 28 global students to intern at National Chung Cheng Univeristy funded by Taiwan Experience Education Program (TEEP), Ministry of Education, Taiwan
- Worked at Advanced Institute of Manufacturing with High-tech Innovations (AIM-HI) – one of leading tech institutions in Taiwan, participating in smart sensing manufacturing research project.

ASEAN in Today's World, Universitas Gadjah Mada & Kyushu University — *Indonesia*
VNU representative — *Feb. 2025*

- Selected as one of four excellent VNU representatives to participate the program funded by Kyushu University, Japan.
- Participated in an ASEAN-focused academic program, strengthening regional understanding, cross-cultural communication, and awareness of science, technology, and society in Southeast Asia.

Temasek Foundation Specialist's Community Action & Leadership Exchange — *Singapore*
VNU Representative — *Feb. 2024*

- Selected as one of 29 VNU students to receive a scholarship for the program.
- Contributed to a group project on youth leadership and sustainability in urban development in Vietnam and Singapore.
- Engaged in a regional leadership and community action program, developing teamwork, communication, and social responsibility in an international environment.

Publications

1. **Long Nguyen Khac** et al. *Adaptive Fuzzy Logic in Backstepping Control for 3-DOF Parallel Robot.* International Conference on Advances in Information and Communication Technology, Springer, 2024. DOI: 10.1007/978-3-031-80943-9_57.
   - ***Main highlights:*** Adaptive fuzzy backstepping control for robust trajectory tracking of a 3-DOF delta parallel robot.
   - ***Presentation:*** Mechatronics and Automation section, The 3rd International Conference ICTA 2024.
2. **Long Nguyen Khac** et al. *YOLOv5s Model with Applied Activation Functions for Personal Protective Equipment Detection in Construction Sites.* Proceedings of the National Conference on Smart Technology Applications in Industry 4.0, Smart City, and Sustainability, 2023.
   - ***Main highlights:*** YOLOv5s-based PPE detection using six activation functions and a dataset of 2,632 construction-site images.
   - ***Presentation:*** National Conference on Smart Technology Applications in Industry 4.0, Smart City, and Sustainability, 2023.

Selected Honors & Awards

Most Popular Poster Award, NTU Singapore Global Connect Fellowship Final Research Poster Presentation	Mar. 2026
NUS ASEAN SPARK Fellow, NUS Enterprise Summer Programme in Entrepreneurship	Jul. 2026
Outstanding Young Face Award, Vietnam National University, International School, awarded in 2024 and 2025	2025 – 2025
TOSHIBA Scholarship, TOSHIBA International Foundation	Sep. 2024
Fighting Spirit Prize, The IEEE Southeast Asia Circuits and Systems Society Hackathon	Dec. 2022
Merit-based University Academic Scholarship, Six-semester recipient for excellent academic performance	2021–2025

Skills

Research	Mathematical modeling, control systems design, autonomous systems (AUV, mobile robots), wireless power transfer, data-driven modeling
Programming	MATLAB, Python, Embedded C/C++, PLC programming
Software	MATLAB/Simulink, SolidWorks, Altium, Proteus, Visual Components, TIA Portal
AI	Data-driven modeling (SINDy, SVD), Machine learning (Random Forest, XGBoost), Computer vision (YOLO)
Languages	Vietnamese (Native), English (Fluent), French (Beginner), Chinese (Beginner)

References

Asst. Prof. Yun Yang — Assistant Professor, School of Electrical & Electronic Engineering, NTU
Yun Yang | yun.yang@ntu.edu.sg

Prof. Her-Terng Yau — Director of AIM-HI, National Chung Cheng University
Her-Terng Yau | htyau@ccu.edu.tw

Assoc. Prof. Nguyen Nhu Tung — Senior Lecturer, Faculty of Engineering and Technology, VNUIS
Nhu-Tung Nguyen | tungnn@vnu.edu.vn

Dr. Le Xuan Hai — Vice Dean, Faculty of Engineering and Technology, VNUIS
Hai-Xuan Le | hailx@vnu.edu.vn